

## CHISINAU INTL, Moldova, Republic of

WMO: 338387

Lat: 46.928N

Lon: 28.931E

Elev: 399

StdP: 14.49

Time Zone: 2.00 (MDA)

Period: 99-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 5.0        | 10.5 | -1.9                          | 5.0 | 6.3  | 3.6 | 6.7 | 12.7 | 23.8                  | 29.0 | 21.6 | 27.9 | 7.2                   | 330  | 0.525 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 22.0                   | 93.2            | 68.1 | 89.7 | 67.6 | 87.4 | 66.5 | 72.0                | 85.0 | 70.2 | 82.8 | 68.9 | 81.2 | 8.6                  | 200  |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 68.1                            | 104.6 | 76.8 | 66.3 | 98.2 | 75.2 | 64.6 | 92.5 | 73.7 | 35.9          | 85.1 | 34.5 | 83.0 | 33.3 | 80.7 | 79.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |       |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| (4) 21.0          | 18.6 | 16.5 | -0.9                       | 97.2 | 6.7                | 3.1 | -5.8                                               | 99.4 | -9.7       | 101.2 | -13.5      | 103.0 | -18.4      | 105.3 |    |    |
| (5)               |      |      | -1.6                       | 74.2 | 6.7                | 2.1 | -6.4                                               | 75.8 | -10.3      | 77.0  | -14.0      | 78.2  | -18.8      | 79.7  |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 51.6   | 28.3  | 31.7  | 41.0  | 52.0  | 62.1  | 69.2  | 73.1  | 72.4  | 62.6  | 51.0  | 42.4  | 32.6  |  |  |
|                                                                        | DBStd       | 17.39  | 9.63  | 9.91  | 8.59  | 6.74  | 6.35  | 5.78  | 5.09  | 5.64  | 6.63  | 7.92  | 8.36  | 8.90  |  |  |
|                                                                        | HDD50       | 2416   | 673   | 512   | 293   | 53    | 2     | 0     | 0     | 0     | 2     | 84    | 256   | 541   |  |  |
|                                                                        | HDD65       | 5613   | 1138  | 932   | 743   | 392   | 131   | 26    | 3     | 7     | 121   | 436   | 679   | 1005  |  |  |
|                                                                        | CDD50       | 3018   | 0     | 0     | 16    | 114   | 378   | 576   | 716   | 693   | 379   | 117   | 28    | 1     |  |  |
|                                                                        | CDD65       | 737    | 0     | 0     | 0     | 3     | 42    | 151   | 255   | 235   | 48    | 3     | 0     | 0     |  |  |
|                                                                        | CDH74       | 8129   | 0     | 0     | 0     | 57    | 540   | 1571  | 2672  | 2659  | 591   | 39    | 0     | 0     |  |  |
|                                                                        | CDH80       | 3124   | 0     | 0     | 0     | 12    | 136   | 534   | 1111  | 1156  | 171   | 4     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 7.5    | 7.9   | 8.2   | 8.9   | 8.1   | 7.4   | 7.4   | 6.9   | 6.5   | 6.8   | 6.9   | 7.7   | 7.8   |  |  |
|                                                                        | PrecAvg     | 20.7   | 1.3   | 1.1   | 1.2   | 1.5   | 2.2   | 2.6   | 2.0   | 2.0   | 1.7   | 1.9   | 1.4   | 1.7   |  |  |
| (15) Precipitation                                                     | PrecMax     | 26.0   | 2.9   | 2.5   | 2.6   | 2.5   | 6.9   | 5.1   | 5.0   | 4.7   | 5.4   | 6.5   | 3.6   | 4.8   |  |  |
|                                                                        | PrecMin     | 13.5   | 0.2   | 0.3   | 0.2   | 0.2   | 0.4   | 0.7   | 0.2   | 0.7   | 0.1   | 0.4   | 0.0   | 0.1   |  |  |
|                                                                        | PrecStd     | 3.7    | 0.7   | 0.6   | 0.6   | 0.6   | 1.4   | 1.2   | 1.0   | 1.1   | 1.3   | 1.4   | 1.0   | 1.1   |  |  |
|                                                                        | PrecStd     | 3.7    | 0.7   | 0.6   | 0.6   | 0.6   | 1.4   | 1.2   | 1.0   | 1.1   | 1.3   | 1.4   | 1.0   | 1.1   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 51.6  | 57.5  | 68.4  | 80.9  | 88.0  | 92.8  | 97.0  | 98.2  | 89.7  | 78.7  | 68.0  | 55.7  |  |  |
|                                                                        |             | MCWB   | 44.2  | 49.5  | 52.6  | 60.2  | 63.5  | 68.3  | 67.3  | 68.3  | 65.6  | 61.8  | 57.6  | 49.4  |  |  |
|                                                                        | 2%          | DB     | 46.4  | 52.0  | 64.1  | 73.6  | 84.0  | 88.2  | 93.1  | 93.4  | 84.5  | 73.4  | 62.3  | 51.6  |  |  |
|                                                                        |             | MCWB   | 42.6  | 44.8  | 50.6  | 57.0  | 63.3  | 67.5  | 68.4  | 68.0  | 62.8  | 59.5  | 54.6  | 47.5  |  |  |
|                                                                        | 5%          | DB     | 43.0  | 47.9  | 58.9  | 69.8  | 80.3  | 86.0  | 89.6  | 89.9  | 80.7  | 68.4  | 57.5  | 48.1  |  |  |
|                                                                        |             | MCWB   | 40.1  | 43.3  | 47.6  | 54.6  | 62.0  | 67.0  | 68.2  | 67.4  | 61.8  | 57.5  | 52.5  | 45.5  |  |  |
|                                                                        | 10%         | DB     | 40.6  | 44.2  | 54.0  | 66.1  | 76.6  | 82.7  | 86.2  | 86.4  | 76.8  | 64.3  | 55.0  | 44.5  |  |  |
|                                                                        |             | MCWB   | 38.5  | 41.1  | 44.8  | 52.9  | 60.8  | 65.1  | 67.3  | 66.2  | 60.7  | 55.5  | 51.0  | 42.1  |  |  |
|                                                                        | 0.4%        | WB     | 45.6  | 50.8  | 53.9  | 60.9  | 68.1  | 74.2  | 73.9  | 73.8  | 68.1  | 64.8  | 58.7  | 51.5  |  |  |
|                                                                        |             | MCDB   | 48.7  | 57.0  | 66.1  | 75.2  | 79.4  | 85.3  | 87.3  | 88.7  | 82.4  | 75.7  | 64.6  | 54.1  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 42.9  | 46.1  | 51.0  | 58.7  | 65.6  | 71.0  | 72.0  | 71.5  | 65.7  | 61.6  | 55.7  | 48.0  |  |  |
|                                                                        |             | MCDB   | 45.3  | 49.9  | 61.7  | 71.3  | 78.2  | 82.5  | 85.1  | 85.4  | 78.2  | 69.6  | 60.4  | 50.2  |  |  |
|                                                                        | 5%          | WB     | 40.5  | 43.3  | 48.7  | 56.4  | 63.7  | 69.2  | 70.4  | 69.6  | 63.9  | 59.0  | 53.7  | 45.2  |  |  |
|                                                                        |             | MCDB   | 42.2  | 46.4  | 57.6  | 66.7  | 75.7  | 80.7  | 83.0  | 82.9  | 75.3  | 66.4  | 57.7  | 47.8  |  |  |
|                                                                        | 10%         | WB     | 38.0  | 40.8  | 46.2  | 54.2  | 61.9  | 67.5  | 69.1  | 68.0  | 62.2  | 56.8  | 51.3  | 41.8  |  |  |
|                                                                        |             | MCDB   | 39.8  | 43.7  | 52.0  | 63.7  | 73.4  | 78.3  | 81.3  | 81.2  | 73.0  | 63.3  | 54.6  | 43.9  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 10.6  | 11.9  | 17.2  | 20.6  | 22.1  | 21.3  | 22.0  | 23.2  | 21.1  | 18.2  | 12.3  | 10.9  |  |  |
|                                                                        |             | MCDBR  | 13.5  | 18.2  | 25.3  | 27.6  | 27.9  | 25.8  | 26.9  | 28.4  | 28.4  | 25.4  | 17.2  | 15.3  |  |  |
|                                                                        |             | MCWBR  | 10.1  | 12.4  | 14.3  | 13.7  | 11.5  | 9.9   | 8.9   | 9.2   | 11.2  | 13.4  | 11.0  | 11.4  |  |  |
|                                                                        |             | MCDBR  | 12.2  | 16.2  | 22.9  | 24.2  | 23.9  | 21.8  | 22.2  | 23.6  | 23.3  | 22.2  | 15.0  | 14.3  |  |  |
|                                                                        | 5% WB       | MCWBR  | 9.6   | 11.8  | 13.5  | 12.7  | 10.9  | 9.5   | 8.9   | 9.0   | 10.7  | 12.5  | 9.9   | 11.1  |  |  |
|                                                                        |             | MCWBR  | 9.6   | 11.8  | 13.5  | 12.7  | 10.9  | 9.5   | 8.9   | 9.0   | 10.7  | 12.5  | 9.9   | 11.1  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.326 | 0.356 | 0.390 | 0.450 | 0.421 | 0.417 | 0.446 | 0.452 | 0.411 | 0.374 | 0.352 | 0.325 |  |  |
|                                                                        | taud        |        | 2.257 | 2.237 | 2.226 | 2.077 | 2.237 | 2.279 | 2.206 | 2.181 | 2.257 | 2.350 | 2.349 | 2.317 |  |  |
|                                                                        | Ebn at Noon |        | 234   | 252   | 261   | 255   | 268   | 270   | 259   | 252   | 252   | 243   | 224   | 221   |  |  |
|                                                                        | Edh at Noon |        | 26    | 32    | 38    | 48    | 43    | 41    | 44    | 43    | 36    | 29    | 23    | 22    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 348   | 585   | 1001  | 1457  | 1882  | 1985  | 1952  | 1770  | 1245  | 788   | 376   | 273   |  |  |
|                                                                        | RadStd      |        | 65    | 93    | 91    | 128   | 161   | 150   | 124   | 106   | 145   | 86    | 50    | 56    |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | +1.08 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +216  | +126  |
| (47) Regional (23 neighbors) |  | +1.17 | N/A     | N/A    | +1.39   | N/A   | N/A   | -209        | -301  | +191  | +113  |

Nomenclature: See separate page